Final project ML

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1. Exploratory Data Analysis

Missing values detection
Following the analysis of the null values, it was surprising to observe the large number of them present in the output, approximately 35 percent. I concluded that the most prudent course of action was to eliminate them. Subsequently, it would be necessary to utilize these data to train various models, thereby necessitating non-inferred data to achieve an accurate estimation.

The same approach was applied to the remaining null values across other categories, as, in comparison to the size of the data, they could be considered negligible.

Outlier Detection
In an initial attempt to analyze the outliers, a raw overview was sought using box plots to obtain a clearer vision of the behavior of our data. Figure 1

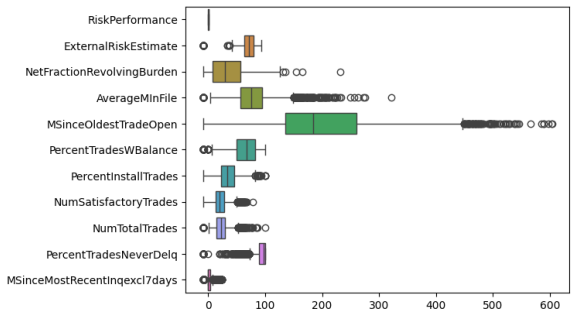

Figure 1: Boxplot of each input variables of the dataset.

In the one hand it can be observed that negative data exist across all categories. Delving further into this matter, it is known that there are approximately 50 such instances in each category, with the exception of MSinceMostRecentInqexcl7days, which has about 130. It has been decided to treat these as NaN values and, therefore, to eliminate these rows. It is noteworthy that the majority are rows of -9, -8 or -7, indicating that they are not widely dispersed and, consequently, will not significantly impact the size of the data set.

Generally, the presence of outliers in financial data can be commonplace due to the natural variability in individuals' financial behavior. Outliers may represent either errors or legitimate atypical cases. This is why it was decided to analyze each category and understand whether the presence of outliers is logical or not.It was concluded that all data appear to be accurate with the exception of those instances where the number of satisfactory trades exceeds the total number of trades, those were dropped.

Class imbalances
After counting the number of observations belonging to each class, it can be confirmed that they are balanced, differing by only 100 units. The observations for non-RiskPerformance are 2344, while those for yes are 2497.

Division and standarization of the data Once we obtain the clean data, and before standardizing anything, we proceed with the separation of the DataFrame into train and test sets after shuffling the data. After this the standarization was done after we perform the train/test split, using only the mean and variance for the training dataset.

Relations between features
To analyze the correlation between the variables, a pairplot has been conducted 2. From this, it can be easily concluded that there are two variables with a very strong correlation, and their plots overlap, allowing us to discard one without affecting the results.

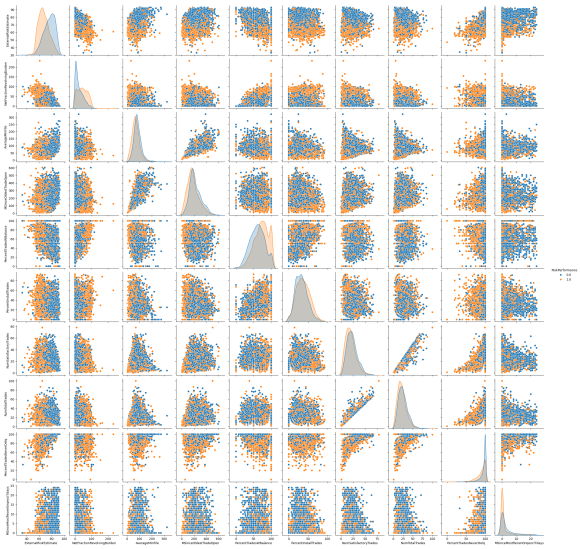

Figure 2: Pairplot.

To decide which variable to eliminate, it was decided to perform a correlation matrix analysis Figure 3. The variable that had a stronger correlation with the target variable would be retained. The results were very similar, -0.13 and -0.14, making it practically indifferent which one to eliminate. However, we opted to remove variable 'NumSatisfactoryTrades', which had the lesser correlation.

2. Classification

2.1 Identification and fitting process of classification models

KNN Classifier

Choosing the correct value for K is crucial for balancing flexibility an the noise in the model. This is why it has been decided to perform cross-validation with the training data to obtain the optimal value of K. Values have been evaluated between 1 and the square root of the number of observations plus 10, as in many cases, the square root of the observations tends to be the optimal value of K.

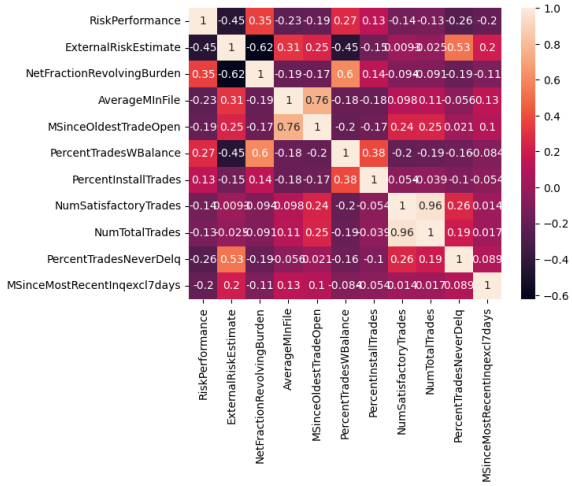

Figure 3: CorrelationMatrix.

As seen in Figure 4, our optimal value is 55 which is near the sqrt(2630) which is the numer of observations.

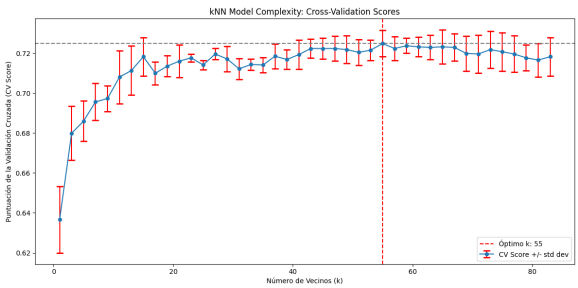

Figure 4: Evolution of accuracies as a function of the number of selected neighbors. The model was evaluated on the test set, and the blue vertical bars represent the standard deviation of the estimator.

The conclusion can be drawn that, as illustrated by the graph, with few neighbors, the model's performance is quite low as it tends to overfit. As the number of neighbors increases, better results are achieved as more information is utilized, but only up to a certain point where it plateaus. The latter occurs because there is likely to be a point where adding more neighbors does not provide significant information for our model.

Logistic Classifier

Despite the fact that the behavior of the variables observed in a pair plot clearly indicated that our data did not conform to a linear model, we sought to verify our assumptions by performing logistic regression. Surprisingly, and getting ahead of what will be discussed later, it was one of the best models.

First and foremost, we conducted a search through cross-validation for the optimal C, which yielded a value of 1. This search also provided an opportunity to observe the behavior of our features and to confirm the observations made in the data analysis section, which was illustrated by the Figure 5.

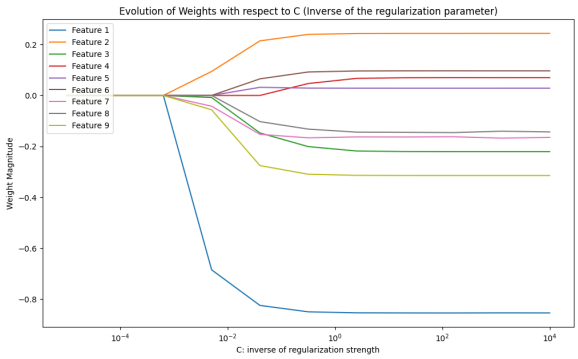

Figure 5: Evolution of the weight of each feature when changing C. Notably, the most significant feature is designated as number 0.1, corresponding to 'ExternalRiskEstimate.' Referring to the correlation matrix generated in the initial section, this feature exhibits the highest correlation with the target variable. Additionally, by examining the pair plot, it is apparent that this feature is the most capable of accurately separating the two classes. All these observations are highly coherent with one another, making us confirn our thoughts about the importance of the feature 1. Furthermore, these insights allow us to think that some variables such as 4 or 5 (AverageMInFile and MSinceOldestTradeOpen), which are among the first to have their weights reduced to zero and exhibit significant correlation with each other, are likely of lesser importance.

After analyzing Lasso and Ridge for regularization, as can be seen in the table, they are practically identical.When comparing the rest of the scores, Lasso wins by a slight margin, and for this reason, it was chosen. I felt that this one was the best because of its capacity of reducing the coefficients of certain features to zero, thus acting as a sort of feature selector. This form of regularization was chosen because Figure 3 in the first section suggested that many of our features might be interrelated.

Table 1: Regularization

Regularization	**Accuracy**	**Precision**	**Recall**	**F1-Score**
Lasso (l1)	0.7286501	0.732644	0.747596	0.739759
Ridge (l2)	0.727740	0.732174	0.744903	0.7383262

Random Forest Classifier

As the third model to be tested, tree-based methods were desired to be used. The analysis began with classification trees, seeking the appropriate depth, which ended up being 6. This value, like the one obtained in random forests, which is also 6, was achieved through cross-validation. This

conclusión can be observed graphically in Figure 6

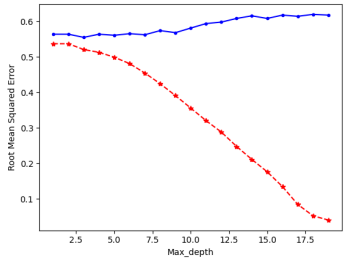

Figure 6: Evolution of the RMSE against the depht of the decision tree. It is worth noting that the obtained graph is entirely coherent. As the tree is allowed to grow larger, the error on the training dataset decreases because the model captures more information from the data. Conversely, the error on the testing dataset also decreases as the model fits better. However, there comes a point where the error on the testing dataset begins to increase, indicating overfitting. The goal is to find the point where both the testing and training errors are minimized.

It is worth noting that in this section, a manual implementation of the model was decided upon for a better understanding of the trees. Despite not reaching the accuracy provided by the 'Random Forest classifier' from the sklearn library, we can assert that this model comes quite close.

On the other hand, taking advantage of the fact that random forest gives us a general idea of the importance of variables in our data, we wanted to verify the results obtained in the first section in a different way. It cam be observed in Figure 7 that the results are very similar to the ones obtained in the correlation matrix in the Figure 3, where the two features with more importance (Riskperformance and NetFractionRevolvingBurden) are the most correlated ones to the output.

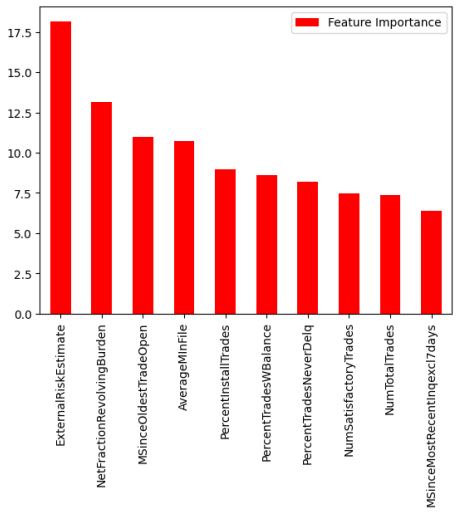

Figure 7: Importance of features

Support Vector Machines

Cointinuing with the implementation of different classification models, it is time for Support Vector Machines model.
The first step, logically, was to search not only for the optimal parameters for our model but also for the type of kernel to use. Initially, I had simply considered, based on the pair plots in Figure 2 such-and-such, attempting to find a kernel that would fit. However, I quickly realized that given the large number of dimensions in the data, the most optimal approach would be to perform cross-validation.

I have decided to analyze both the linear and radial kernels. Ultimately, I have opted for the radial kernel. This decision is based not only on the higher accuracy it provides but also on the observations from the pairplot in the Figure 2, which reveal that the majority of the graphs indicate a significant overlap of the classes. Additionally, there is a noticeable presence of outliers, suggesting that the flexibility of the RBF kernel will lead to better classification performance. The optimal params for the Rbf kernel obtained by cross-validation are: C=1, gamma=0.1. Wen can see this graphically in the Figure 8 where with high values of gamma, each group of values affect only to a small region of the decision which means that the model becomes really sensitive to the training data what end in overfitting (red line) . In this scenario, the types of kernels that will be created with low gamma values will have irregular shapes. Conversely, with high gamma values, generalization is prioritized, and therefore these kernels will be more regular (as indicated by the blue line). Regarding the influence of C, it simply affects the number of training points that can be misclassified. In our case, as it was said before, the optimal value of C is 1 which is a value not too strict and not too lenient.

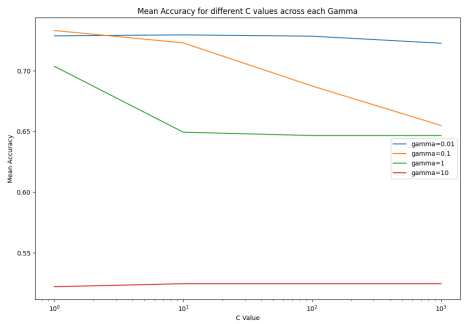

Figure 8: Evolution of the params gamma and C in a rbf kernel

Neural Networks

To conclude the analysis of the behavior of different classification models with our data, it was decided to implement a 'Neural Network' model.

Once again, cross-validation has been employed to obtain the optimal parameters. In this case, it was decided to use two hidden layers, and finally, after cross-validation, these two layers have the dimensions (10, 10) with a learning rate of 0.0001. Additionally, due to the binary output, it was believed that the most convenient approach was to use a logistic activation function. After cross validation our thoughts were confirmed.

I would like to emphasize the surprising nature of the results; despite the effectiveness of neural networks, they did not yield remarkable outcomes in this case. The conclusion drawn is that, given our relatively small dataset, neural networks may be more sensitive to data variability and overfitting. Therefore, it stands to reason that simpler models may generalize better in limited datasets.Hence, the computational cost associated with neural networks may not be justifiable for our dataset

2.2 Comparative analysis of the fitted models

Among all the models that have been evaluated, as shown in Table 2, we can observe that they all resemble each other closely. Therefore, in order to achieve a better comparison as well as to make a decision, it has been decided to perform the ROC curves and the calibration curves which can be seen in Figure 9 and in 10.

Table 2: Comparación de modelos

Modelo	Métricas de Desempeño			
	Accuracy	Precisión	Recall	F1-Score
KNN	0.724793	0.729220	0.743857	0.0.736018
Logistic Regression	0.7286501	0.732644	0.747596	0.739759
Classification Trees	0.703581	0.697719	0.754015	0.723591
Random Forest (sk)	0.723416	0.717047	0.767879	0.741161
Random Forest (hand)	0.636664	0.663102	0.597111	0.628378
Rbf	0.733058	0.728212	0.772145	0.749127
Neural Networks	0.724793	0.725253	0.752405	0.738425

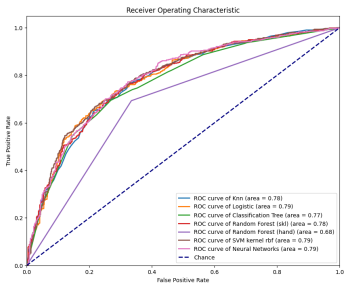

Figure 9: Comparison of the ROC curves of all the models presented. We can observe that the results are all very similar, if not practically identical. It's obvious that the manually implemented random forest model would give somewhat inferior results. This means that we need to focus in other things in order to determine which are the best ones.

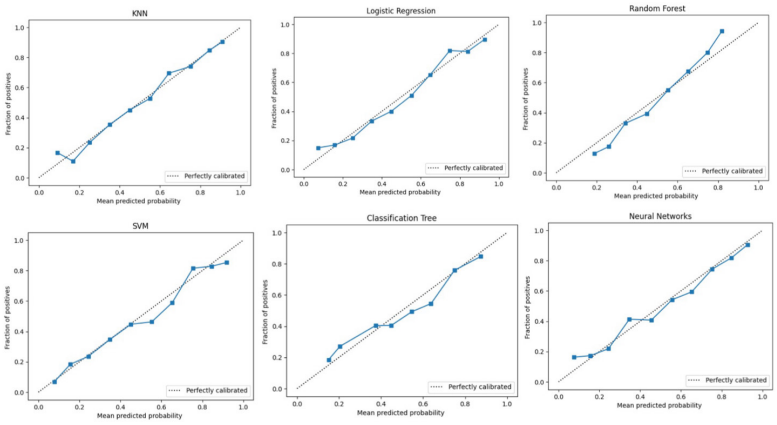

Figure 10: Logistic Regression and Neural Networks appear to be the most well-calibrated models according to these graphs, with KNN also exhibiting good performance. Random Forest displays a slight overestimation at higher probabilities, whereas SVM and Classification Tree demonstrate an underestimation of probabilities.

3. Unsupervised Learning

In this section, we will proceed with the implementation and comparison of unsupervised learning models. Specifically, we will explore two methods: Principal Component Analysis (PCA) and clustering.

3.1 PCA fitting process and analysis

PCA is a supervised learning model that helps us gather more information about our features by reducing dimensionality. To do this, different PCs were created and the data was previously standarized. In particular, considering our data and the threshold stablished (0.9), the optimal number of PCs was 6, even though, after, it will be explained that even 4 PCs work well . In the Figure 11, it can be observed that depending on the threshold set (in our case it was 0.9), which represents the amount of variance to explain, one number or another of PCs will be chosen.

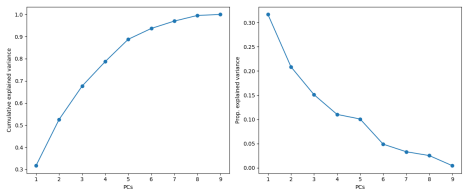

Figure 11: Left: a scree plot depicting the proportion of variance explained by each of the 9 principal components. Using the Elbow method it can be seen what was said before, the optimal numer of PC is aroud 6,it is the point where the increase in explained variance from adding another principal component becomes less pronounced. Right: the cumulative proportion of variance explained by the four principal components.

After analyzing the model, it was desired to observe how the weights associated with each feature

vary when creating the three main principal components which can be seen the analysis in 12.

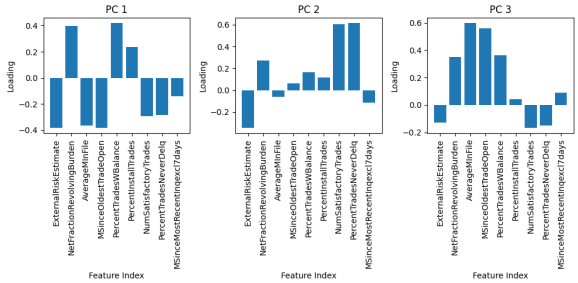

Figure 12: Comparative of the first 3 PCs weights. As we can see, 'External risk' plays a significant role in the first two, corroborating what we have been stating throughout the document. It is evident that while in the first principal component the weights are quite distributed among all features, the second suggests that individuals with a low 'ExternalRiskEstimate' and high 'NumSatisfactoryTrades' and 'PercentTradesNeverDel' are likely to belong to the same group regardless of the other features."

Once our new data with reduced dimensions has been obtained, it has been decided to compare the results obtained after training SVM and logistic regression with these, as well as changing the number of PCs used, firts with the optimal (6) and also with 4. In both cases, it can be observed in Table 3 that, as expected, the scores decrease, since PCA removes information. However, it allows us to work with much simpler data, thereby reducing computational costs. Although we are not currently dealing with large amounts of data and the differences are not so noticeable, it is likely that with larger datasets, dimensionality reduction will be indispensable.

Table 3: PCA model's comparative

Modelo	Metrics			
	Accuracy	Precisión	Recall	F1-Score
Logistic Regression	0.7286501	0.732644	0.747596	0.739759
Logistic Regression (PCA,PCs=6)	0.727740	0.732174	0.744903	0.7383262
Logistic Regression (PCA,PCs=4)	0.7114209	0.715032	0.732889	0.7236978
Rbf	0.733058	0.728212	0.772145	0.749127
Rbf (PCA,PCs=6)	0.7287725268	0.7261916	0.761722645	0.74334
Rbf (PCA,PCs=4)	0.702953	0.71158	0.713259	0.712303

To conclude with PCA, I wanted to show how our dataset looks when reduced to two and three dimensions in the Figure 13

3.2 Identification and fitting process of clustering algorithms

In order to discern similarities and patterns within the data, it has been resolved to employ clustering.

First it has been implemented an hierarchically clustering that can be seen in Figure 14. In the image it can be seen that the optimal number of clusters to utilize is two since it is the largest band

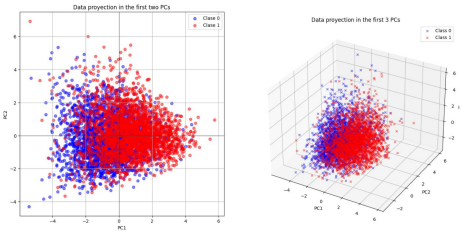

Figure 13: Left:Data in two dimensions. Right: Data in three dimensions.We can observe that with two dimensions, the separation appears somewhat constrained, as despite the initial visual separation of the two classes, they overlap. When represented in three dimensions, we can see that with, for example, a separation hyperplane, they would separate quite well.

in the representation. However, this is not related to the output of classes being binary; rather, it is merely that in this specific instance, two clusters more effectively elucidate the data.

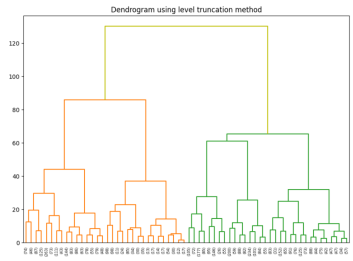

Figure 14: Dendrogram obtained from hierarchically clustering representing the possible cuts in the data for the creation of clusters. The term level truncation refers to the limitation of the tree's growth.

On the other hand, the Silhouette method has been utilized to determine how well each object has been classified upon being assigned to a cluster. As can be observed in the Figure 15, the average of the method is 0.34181515677100466 when considering 2 clusters, which suggests that many points might have been better suited to a different cluster, as the result is quite close to 0. Nevertheless, since it is not negative, it does not confirm that these points are misclassified. Moreover, this is the best value obtained after calculating these values for different numbers of clusters, as seen in the Table 4.

Table 4: Average silhouette scores comparative

N clusters	Silhouette scores
2	0.34181515677100466
3	0.3364262132226679
4	0.283600691666092
5	0.27817006282472195
6	0.2666242763775426

4. Conclusions

4.1 Some questions...

1. Are the hypothesis about variable importance from Part 1 confirmed in Part 2? The response is a definitive yes. Throughout the document, various methods have been employed that provide information on the importance of the features, always referencing previous analyses. This can be observed in figures such as the pairplot in Figure 2, or the bar chart in Figure 7, even in the evolution of the weights in Figure 5. Each consistently leads to the same conclusion, that 'ExternalRiskPerformance' and 'NetFractionRevolvingBurdeon' are the most important features while others as 'PercentInstallTrades' or 'NumSatisfactoryTrades' are not as important.

2. Which is the best classification algorithm? Why? This question is somewhat difficult to answer, as seen in the Table 2, because most of them exhibit similar performance. However, I would primarily lean towards two options: Logistic Regression or SVM using a radial kernel. Logistic Regression is easy to implement, interpret, and very efficient to train, and it also performs very well with small datasets. SVM excels in high-dimensional spaces and is less prone to overfitting. Ultimately, I would opt for Logistic Regression based on its computational cost, which is very low compared to SVM as well as the calibration curve seen in Figure 10

3. If the clustering methods are used for the classification task, would the performance be similar to the algorithms in Part 2? We can compare both methods by examining the Silhouette score in clustering and the F1-scores in classification algorithms, as both achieve a balance between how the methods capture positives (cohesion within a cluster) and ensure accurate categorization (separation between clusters). By scaling our 0.34181515677100466 value from the Silhouette score to a range of 0, 1, just as the F1-score is, we find it to be approximately between 60 to 70. Therefore, we can conclude that both exhibit similar performance.

4. How do we ascertain the relative importance of precision, recall, and the F1-score for our specific problem? Precision is a crucial measure when the cost of a false positive is high, thereby unfairly denying credit. If the focus is more on monetary losses, recall would be the measure of choice, as failing to identify false negatives leads to financial losses. Finally, if one seeks a balance, especially in a financial environment where both excess and deficit are negative, the F1-score is the best measure.

Extra As an optional part, I chose to base my work on the last topic of the book 'An Introduction to Statistical Learning' titled 'Multiple Testing'. Essentially, what is discussed in the topic is that multiple testing is crucial to ensure that our statistical conclusions are not the result of chance. It has been used cross validation to obtain multiple accuracy estimates for two classification models (SVM and Logistic Regression), enhancing the reliability of performance comparisons. It uses a t-test (learned last year) to statistically determine if there is a significant difference in accuracy between the models. By doing so, it ensures that our conclusions about model performance are robust and not due to random chance. The p-values should ideally be below a predefined significance level, commonly set at 0.05 or 0.01. This would suggest that the observed differences in model accuracy are statistically significant and unlikely to have occurred by chance.

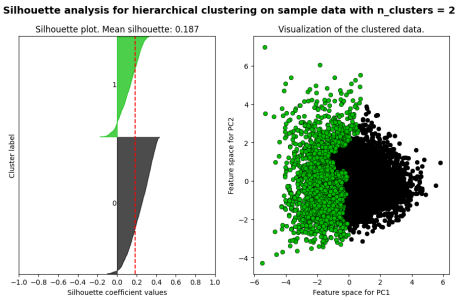

Figure 15: Dendrogram obtained from hierarchically clustering